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(1) Research Program/Project/Study Title: VisionVoice: An Image-to-Speech Application for Visually Impaired Individuals	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	
Component Leader:	
Academic Rank:	

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/Biological Sciences ☐ Food Research/ ☒ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster	(3) Research Center [Check the concerned center] ☒ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☐ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center
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(4) Site of implementation			
		Municipality/City:	Pontevedra
		Province:	Capiz
		Region:	Region VI
(5) Classification		(6) Mode of Implementation	
Research:	Development:		
☒ Basic	☒ Pilot Testing	☐ Single Authorship/Proponent	
☐ Applied	☐ Tech. Promotion/Commercialization	☒ Collaborative Research (within campus)	
		☐ Collaborative Research (intercampus)	
		☐ Collaborative Research (external agency)	

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(7) Sector/Commodity Education Sector	(8) Discipline/Field of Specialization Computer Science
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(9) Rationale

VisionVoice: An Image-to-Speech Application for visually impaired people, revolves around its transformative impact on users' accessibility, independence, and inclusion. VisionVoice converts visual information into spoken words, allowing users to independently access a diverse range of visual content while fostering inclusion and equal access to information. It gives people the ability to navigate their surroundings, absorb visual content, and communicate effectively, promoting independence, confidence, and social inclusion. VisionVoice also improves safety and mobility by delivering critical information about the environment, resulting in an improved quality of life for visually impaired people. VisionVoice strives to remain a valued tool by continuously improving and innovating, incorporating cutting-edge technology to meet the different needs and preferences of users. Overall, VisionVoice represents an important improvement in assistive technology.

(9-A) Project Description

VisionVoice is a mobile application that helps vision impaired people by giving real-time aural descriptions of their environment using picture recognition technology. It recognizes objects and text using powerful algorithms and converts them into clear voice. Text recognition, offline mode, adjustable preferences, voice control, and accessibility options are among the key features. The app encourages independence, access to visual information, and inclusivity by providing real-time assistance in navigating the environment more confidently.

(10) Objectives/Statement of the Problem (for Social Sciences) (Statement of the specific purpose to address the problem areas of the project.)

Objectives of the Study

General Objectives:

This study is aims to design and develop a VisionVoice: An Image-to-Speech Application for Visually Impaired Individuals.

Specific Objectives:

Specifically, this study aim to:

1. Design and develop a VisionVoice: An Image-to-Speech Application for Visually Impaired Individuals to develop a user-friendly and efficient application that enables visually impaired individuals to:
 - a. Independently access and understand visual content in various formats, including printed text, signs, labels, documents, and scenes.
 - b. Navigate their surroundings safely and efficiently by receiving real-time audio descriptions of their environment.
 - c. Communicate effectively with sighted individuals by providing accurate descriptions of

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visual elements and facilitating smoother interactions in various contexts.

- d. Enhance educational and employment opportunities by enabling access to visual learning materials and productivity tools.
- e. Foster inclusivity and equality by ensuring equal access to information, entertainment, and services for visually impaired individuals.

(11) Review of Literature/Prior Art Search

Related Studies

Third Eye: Object Recognition and Speech Generation for Visually Impaired

Detecting and recognizing the objects and generating speech about the objects helps visually impaired in a great way in understanding their surroundings. Required a mechanism to assist the visually impaired person to travel independently with the ability to identify objects in their path, and the ability to generate speech describing the objects detected in the scene. This can be achieved with the help of YOLOv5 image detection model and text to speech converters such as gTTS and pyttsx3 modules in python. The proposed method called Third Eye, giving better accuracy in detection and speech generation to help the visually impaired people Compared to existing approaches. YOLO v5 is trained on custom dataset of 15 objects along with MS COCO 2017 Dataset of 80 objects (95 objects overall). The output labels of the model are transformed to text and later converted to audio format and are presented to the visually impaired, through a speaker. We compared two python libraries for audio conversion, one is pyttsx3, and the other is gTTS.

Artificial intelligence for visually impaired

The eyes are an essential tool for human observation and perception of the world, helping people to perform their tasks. Visual impairment causes many inconveniences in the lives of visually impaired people. Therefore, it is necessary to focus on the needs of the visually impaired community. Researchers work from different angles to help visually impaired people live normal lives. The advent of the digital age has profoundly changed the lives of the visually impaired community, making life more convenient. Deep learning, as a promising technology, is also expected to improve the lives of visually impaired people. It is increasingly being used in the diagnosis of eye diseases and the development of visual aids. The earlier accurate diagnosis of the eye disease by the doctor, the sooner the patient can receive the appropriate treatment and the better chances of a cure. This paper summarises recent research on the development of artificial intelligence-based eye disease diagnosis and visual aids. The research is divided according to the purpose of the study into deep learning methods applied in diagnosing eye diseases and smart devices to help visually impaired people in their daily lives. Finally, a summary is given of the directions in which artificial intelligence may be able to assist the visually impaired in the future. In addition, this overview provides some knowledge about deep learning for beginners. We hope this paper will inspire future work on the subjects.

Design and Implementation of Text To Speech Conversion for Visually Impaired People

A Text-to-speech synthesizer is an application that converts text into spoken word, by analyzing and processing the text using Natural Language Processing (NLP) and then using Digital Signal Processing (DSP) technology to convert this processed text into synthesized speech representation of the text. Here, we developed a useful text-to-speech synthesizer in the form of a simple application that converts inputted text into synthesized speech and reads out to the user which can then be saved as an mp3.file. The development of a text to speech synthesizer will be of great help to people with visual impairment and make making through large volume of text easier.

Perceptors: A Real Time Object Detection System with Voice Feedback and Distance Approximation for Blind

Engineering solutions have impacted everyone's lives, but those designed to aid people with disabilities have been especially significant. However, the current pricing of modern assistive gadgets

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does not match the market's needs. In its latest study, WHO puts the number of visually impaired people at 285 million. Of these, 246 million are blind and an estimated 39 million are visually impaired. Blind people face many problems and difficulties in business and social activities. It also has the potential to contribute to society if the opportunity arises. In this paper we propose to you "Perceptors"—An A.I. powered smart glasses that can detect objects in real time and alert users when objects are nearby. This research is an attempt to give visually impaired people a chance to experience a moderately normal life by warning users in advance of approaching objects and helps users identify objects in real time.

Enhanced portable text to speech converter for visually impaired

The handwritten text reader is designed to help the visually impaired listen to an audio read-back of printed and handwritten scanned text. A hand-held page scanner is used to scan the text to be read. The image from the scanner is sent to the application in the paired Android phone over Bluetooth. An open source optical character recognition OCR engine, Tesseract is used to extract the text from the image, and this extracted text is converted to speech. Tesseract OCR engine is further trained to recognise handwritten text for a specific user. This OCR engine is trained with handwritten datasets. In addition to English, the application supports two regional languages-Hindi and Bengali.

(12) Scientific Basis/Theoretical Framework (Include concepts, theories, and variables that indicate relevant relationships to be tested.)

The VisionVoice: An Image-to-Speech Application for Visually Impaired Individuals is based on the following theoretical frameworks:

User-Centered Design (UCD):

UCD principles emphasize the importance of involving end-users in the design process. Understanding the needs, preferences, and challenges of visually impaired individuals guides the development of VisionVoice, ensuring it meets their specific requirements and enhances usability.

Accessibility and Inclusive Design:

Drawing from the principles of inclusive design, VisionVoice should prioritize accessibility. This includes considerations for diverse user abilities, providing alternative modes of interaction, and ensuring compatibility with assistive technologies.

Human-Computer Interaction (HCI):

HCI theories and models contribute to the design of the user interface and the overall interaction experience. Concepts such as feedback loops, affordances, and mental models can be applied to create an intuitive and efficient user interface for VisionVoice.

Speech Synthesis Models:

Leveraging theories from the field of speech synthesis, VisionVoice can be built on models like concatenative synthesis or formant synthesis. Understanding the intricacies of generating natural-sounding speech is crucial for the application's success.

Natural Language Processing (NLP):

NLP theories guide the language processing capabilities of VisionVoice. Techniques from deep learning models, such as Transformers, can be employed to improve language understanding, context

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awareness, and the generation of coherent and contextually relevant speech.

Psycholinguistics:

Insights from psycholinguistics can inform the prosody and intonation patterns used in VisionVoice. Understanding how humans process and interpret spoken language helps create a more natural and engaging auditory experience for users.

Acoustic and Auditory Perception:

Theoretical concepts in acoustic and auditory perception contribute to the design of VisionVoice's audio output. Considerations for pitch, speed, and clarity are essential to ensure that the synthesized speech is comfortable and easily comprehensible for visually impaired users.

Ethical AI and Privacy:

Embedding principles of ethical AI ensures that VisionVoice respects user privacy, maintains data security, and avoids potential biases. Ethical considerations should be integral to the development, deployment, and usage of the TTS application.

Continuous Improvement and Adaptability:

Drawing from agile development methodologies, the theoretical framework should support continuous improvement. Regular user feedback, updates based on technological advancements, and adaptability to emerging needs ensure that VisionVoice remains effective and relevant over time.

Empowerment and Independence:

The theoretical underpinning of VisionVoice should be grounded in empowering visually impaired individuals, fostering independence in accessing information, and promoting social inclusion.

(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layout; c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

Materials

The study used physical environment in the study which includes hardware and software. The following has its specification as follows:

- Laptop or Personal Computer
- Mobile Phone

Hardware Specifications

Hardware	Minimum Requirements
Computer or Laptop	Core i7
Mobile Phone	Android 8.0+

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Memory (RAM)

At least 4GB

Software Specification

Software

Definition

For Development:

- Python and Java
- Visual Studio Code

a multi-platform, object-oriented, and network-centric language that can be used as a platform in itself. It is a fast, secure, reliable programming language for coding everything from mobile apps and enterprise software to big data applications and server-side technologies.

a code editor redefined and optimized for building and debugging modern web and cloud applications.

Device Compatability

- Android 7+
- IOS

SOFTWARE DESIGN MODEL

Software Development Life Cycle (SDLC)

Iterative Development Model

In Iterative Model, iterative process starts with a simple implementation of a small set of the software requirements and iteratively enhances the involving version until the complete system is implemented and ready to be deployed. At each iteration, designed modifications are made and new functional capabilities are added. The basic idea behind these methods is to develop a system through repeated cycles and in smaller portion at a time.

Below is the visualization of the Iterative Model to be used in this study.

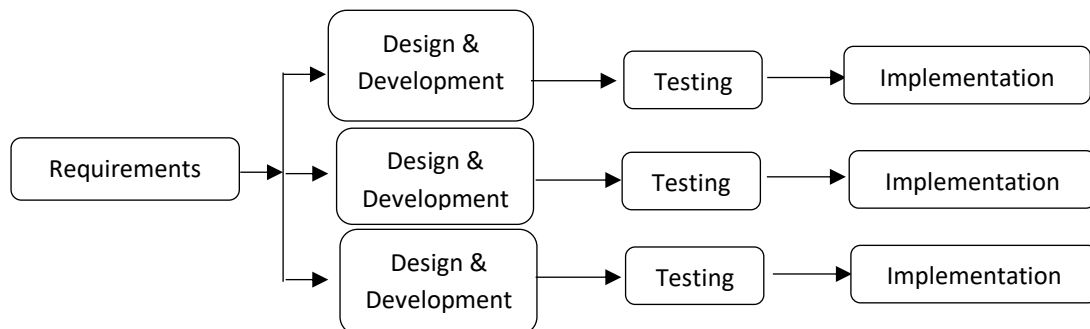


Figure 1. Iterative Model of the VisionVoice: An Image-to-Speech Application for Visually Impaired Individuals

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Conceptual Framework

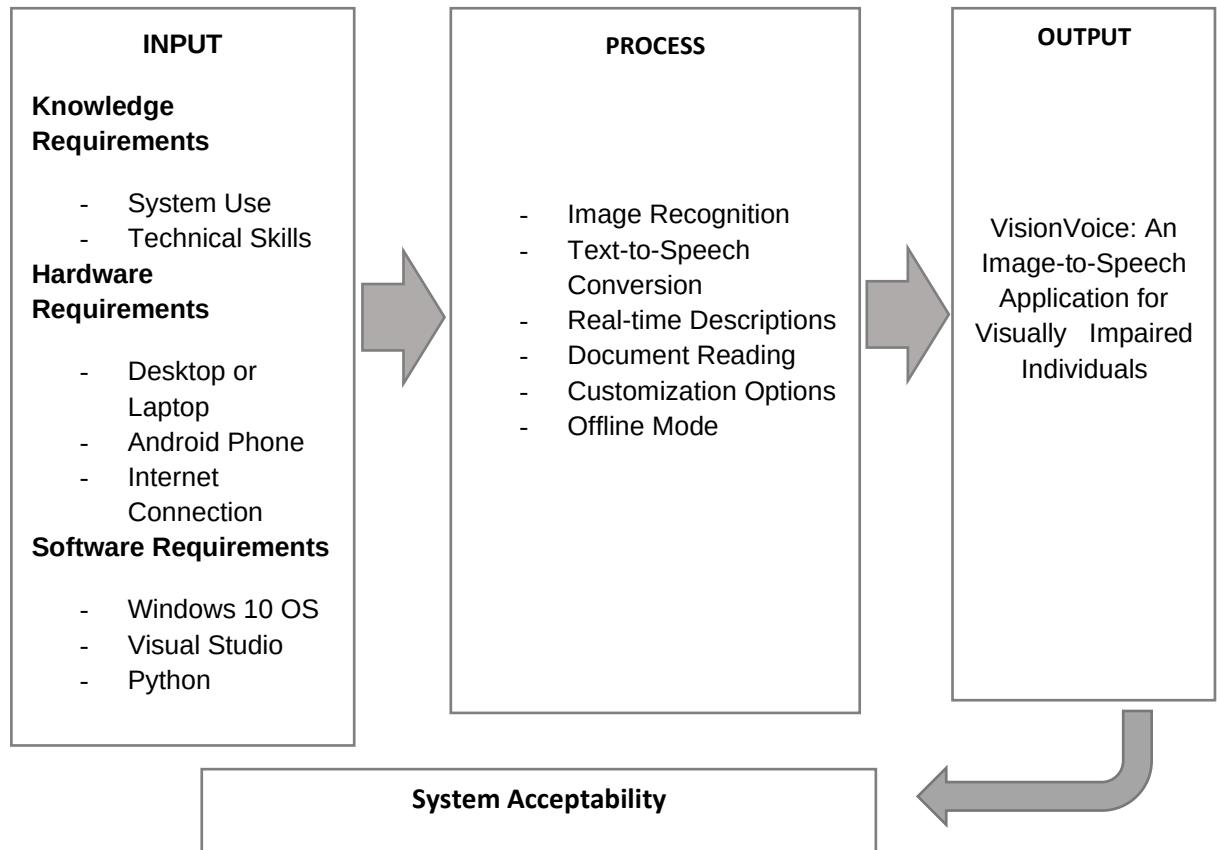


Figure 2. Conceptual Framework of the study

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(14) Major Activities/Work plan (Enumerate in chronological order the task to be undertaken. Use gantt chart.

TASK NAME	MONTHS				
	2023				
	March	April	May	June	July
Design the system to be developed					
Develop the Project Plan					
System and Software Design					
Design the system					
Design the architecture					
Building the database and the program					
Implementation and Unit Testing					
Building of data files/codes					
Check the system and debug for errors					
Integration and System Testing					
Check if the system is user friendly					
Testing as a complete					

(15) Expected Output (Indicate the specific products, processes or services which the project is expected to produce; quantify when possible.)

The expected output of the VisionVoice Image-to-Speech application for visually impaired individuals

Specific expected outputs:

1. Image Recognition: Advanced image recognition technology will analyze images captured by the device's camera and identify text, objects, and scenes.
2. Text-to-Speech Conversion: The application will convert the recognized text into spoken words using high-quality text-to-speech synthesis, making it accessible to users through audio output.
3. Real-time Description: Users will receive real-time audio descriptions surroundings, including text, objects, people, and environmental elements, enabling them to navigate and interact with their environment independently.
4. Document Reading: VisionVoice will allow users to capture and read printed documents, books, labels, and signs by converting the text into speech, providing access to a wide range of written

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materials.

5. Customization Options: The application will offer customization options, allowing users to adjust speech rate, volume, language, and other settings to suit their preferences and needs.
6. Offline Mode: VisionVoice will support offline mode, enabling users to access essential features and functionalities without an internet connection, ensuring reliability and accessibility in various environments.

(15) Technology Transfer Activity Design (Indicate the specific products, activity design, processes or services to be transferred as extension activities)

A technology transfer activity design of the VisionVoice Image-to-Speech Application for Visually Impaired Individuals as an extension activity of the campus could be designed as follows:

1. Development of the VisionVoice: Based on the identified needs, design and develop the VisionVoice: Text-to-Speech Application, this could involve hiring IT professionals and web developers to enhance the website and the necessary features.
2. Training and capacity building VisionVoice: Image-to-Speech Application, has been developed, conduct training and capacity-building activities for university staff, including the IT team, department heads, and front-line staff. This will ensure that they have the necessary skills to operate and maintain the application.
3. Pilot testing: Conduct a pilot test of the VisionVoice: Image-to-Speech Application, to identify any glitches or issues that need to be addressed. This can be done by selecting a small group of students to use the system and provide feedback.
4. Launch and dissemination: Once the VisionVoice: Image-to-Speech Application, has been fully developed and tested, launch it to the entire university community. This can be done through an official launch event, promotional materials, and targeted communication to different departments and units.
5. Maintenance and updates: Ensure that the VisionVoice: Image-to-Speech Application, is regularly maintained and updated to keep it running smoothly and efficiently. This will involve regular checks for bugs, updates to security features, and improvements to functionality based on user feedback.
6. Evaluation and feedback: Finally, conduct an evaluation of the VisionVoice: Image-to-Speech Application, to assess its impact and effectiveness. This can be done through surveys, focus group discussions, and usage data analysis. Use this feedback to improve the system further and to inform future technology transfer activities.

(16) Targets Beneficiaries (Who the clientele are and what are the expected outcome/effects of the use of the projects output)

This study will be beneficial to the following:

1. **Visually Impaired Individuals** - People with varying degrees of visual impairment, including those with partial sight or low vision, who may benefit from assistive technologies to enhance their accessibility.
2. **Elderly Individuals with Age-Related Vision Loss** - As age-related vision impairments become more common with advancing age, elderly individuals who experience visual decline could find VisionVoice helpful in maintaining independence and accessing information.
3. **Students with Visual Impairments** - Students who are visually impaired may benefit from VisionVoice in educational settings, helping them access visual content in

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textbooks, presentations, and other learning materials.

4. **General Public with Temporary Vision Loss** - Individuals with temporary vision loss due to injury, surgery, or other factors may also find temporary use of VisionVoice helpful in maintaining productivity and independence during their recovery period.

(17) Personnel Requirement

Name	Role (Program, Project, Study Leader)	Field of Specialization
Michael John D. Moneva	Project Leader	Computer Science
Venus B. Dorado	Member	Computer Science
Roden B. Pelegrino Jr.	Member	Information Technology
Shawn A. Dacles	Member	Computer Engineering
Virginia G. Figueroa	Member	Computer Science

Percent Time Devoted to the Program/ Project/ Study

Program/Project Leader (in hours): 80 hours

Program/Project Members (in hours): 40 hours

(18) Budgetary Requirement

SOURCES OF EXPENSES	Quantity	Amount (PhP)
A. Personnel Services (For studies needing Research Assistants, Informant/s, Guide)	1	₱20,000
B. Maintenance and Other Operating Expenses (MOOE) (modify and include only whichever is applicable)		
- Transportation allowance		₱1,000
- Meal allowance during data gathering		₱2,000
- Professional fees (laboratory analysis and other related fees)		₱2,500
- Miscellaneous (for incidental expenses)		₱1,000
- Printing		₱500
C. CAPITAL OUTLAY		₱7,000
Total		₱34,000

(19) Literature Cited (Include only those references published in reputable journals, and at least that from CAPSU Research Journal, <http://researchjournal.capsu.edu.ph/>)

Guravaiah, K., Bhavadeesh, Y. S., Shwejan, P., Vardhan, A. H., & Lavanya, S. (2023, January 1). *Third Eye: Object Recognition and Speech Generation for Visually Impaired*. Procedia Computer Science. <https://doi.org/10.1016/j.procs.2023.01.093>

Wang, J., Wang, S., & Zhang, Y. (2023, April 1). *Artificial intelligence for visually impaired*. Displays. <https://doi.org/10.1016/j.displa.2023.102391>

Isewon, I., Oyelade, J., & Oladipupo, O. O. (2014, April 5). *Design and Implementation of Text To Speech Conversion for Visually Impaired People*. International Journal of Applied Information Systems. <https://doi.org/10.5120/ijais14-451143>

Bharadwaj, R., Sonawane, H., Patil, M., Patil, S., & Jadhav, V. (2023, January 1). *Perceptors: A Real Time Object Detection System with Voice Feedback and Distance Approximation for Blind*.

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Enhanced portable text to speech converter for visually impaired | International Journal of Intelligent Systems Technologies and Applications. (n.d.). International Journal of Intelligent Systems Technologies and Applications. <https://doi.org/10.5555/3271948.3271951>

(20) Capsule Curriculum Vitae (One-page brief C.V. only)

MICHAEL JOHN D. MONEVA



CONTACT

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Languages Hobbies

English	Watching
Filipino	Coding
Hiligaynon	

PERSONAL DATA

Date of Birth	: November 12, 1997
Place of Birth	: Ibaca, Pres. Roxas, Capiz
Age	: 26 years old
Height	: 5'4"
Weight	: 62 kg
Civil Status	: Single
Gender	: Male
Religion	: Roman Catholic
Citizenship	: Filipino

EDUCATIONAL ATTAINMENT

Tertiary	: Capiz State University Pontevedra Bachelor of Science in Computer Science 2021
Secondary	: Elizalde Academy President Roxas, Capiz 2014
Elementary	: Pres. Roxas East Elementary School President Roxas, Capiz 2010

SKILLS AND QUALIFICATION

- Web designing and developing using C#.
- Web designing and developing using WordPress as CMS.
- Graphics Designing